

Adrien Fabre

Adapted by Steve Jamson

HEALING THE WORLD

A Plan to End Climate Change and Extreme Poverty

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Preface, by *Gabriel Zucman*

How can we move towards a fairer world while effectively combating climate change? This book proposes a clear and convincing solution that deserves to be widely debated.

The task is not an easy one. There are countless proposals to reduce greenhouse gas emissions, the inequalities of which are obvious. This failure to move forward together on the two major challenges of the 21st century – combating climate change on the one hand, reducing economic disparities on the other – has, for several decades, been at the heart of our collective inability to preserve the climate.

Adrien Fabre's book is invaluable in demonstrating that this tension can be overcome.

The most reliable instrument for tackling climate change is well known: it consists of capping greenhouse gas emissions through a system of tradable quotas, the quantities of which must gradually decrease to reach zero shortly after the middle of the 21st century.

By redistributing the revenues generated equally among all human beings, i.e. by allocating the same emissions permit to each inhabitant of the planet, a criterion of justice that is difficult to contest, this mechanism would make it possible to finance a universal basic income of around €50 per month per person between 2030 and 2060, thus eradicating the most extreme forms of poverty.

A utopia? Since completing his Ph.D., Adrien Fabre has

specialized in opinion surveys on climate and redistribution. And this is where his treatise becomes fascinating. For his meticulous work overturns a commonly accepted idea: that people in wealthy countries are hostile to international redistribution. “In reality, he writes, people are willing to embrace ecological change and solidarity, provided that the effort is international and equitably shared, and that it falls first and foremost on the wealthiest.”

It is not democracy that stands in the way of progress; it is the conservatism of economic elites and defeatism, for which this book is an indispensable antidote.

In fact, international transfers, although highly insufficient, are far from being completely negligible today: on the order of 0.4 points of GDP for a country like France, or over 10 billion euros a year. There is nothing unrealistic about envisaging a doubling or tripling of these flows in the years to come. In fact, OECD countries have pledged to increase development aid to 0.7 points of GDP. The plan proposed by Adrien Fabre would result in international transfers of around 0.6% of global GDP, and would be in line with this trend.

Proof of the urgency of the subject; initiatives to reconcile international economic justice and climate sustainability are multiplying. Nobel Prize-winning economist Esther Duflo estimates that the countries of the North should pay \$500 billion a year to poor countries (0.5% of global GDP), just to compensate for the loss of human life caused by the current emissions of people living in industrialized countries. At the G20 Finance Ministers’ meeting in São Paulo in February 2024, I presented a proposal for a coordinated minimum tax on the world’s billionaires, which would, with a modest minimum rate of 2 percent on the wealth of the individuals concerned, raise at least \$250 billion a year. It would be perfectly logical to allocate at least part of this revenue to the poorest countries, which, although home to relatively few billionaires, have made a powerful contribution to the enrichment of those in the North by giving them

access to their markets.

Adrien Fabre's book, based on the most recent research and his pioneering work, adds to this vital debate on the future of our planet. It is essential reading for all citizens committed to justice and progress.

Gabriel Zucman

Professor at the Paris School of Economics and UC Berkeley

April 15, 2024

To the climate movement,

Motivation

Imagine a peaceful society, where everyone lives in dignity, where the conditions necessary for well-being are sustainably guaranteed for all humankind; this is the world we want to live in. Why am I convinced that this book can help humanity evolve in this direction? Because after ten years of studying economics and climate change, after completing a Ph.D. on the conditions for the feasibility and acceptance of ecological transition, and focusing my academic research on opinions about climate and redistribution, my colleagues and I made an astounding discovery, full of hope for the future. There is a solution to end both global warming and extreme poverty, recognized by all experts and supported by a majority worldwide. Although identified as early as 1990, this solution has not been discussed since then. Indeed, it implies major North-to-South transfers of wealth, which the countries of the North have refused to do since the first climate negotiations. But no one had consulted with the populations of these countries to find out what they wanted. Yet, by conducting representative surveys in 20 countries that account for three-quarters of the world's greenhouse gas emissions, it appears that this global solution to climate change would in fact be widely supported, even in the Global North. This promising result motivated me to bring this solution to life by writing this book and co-founding an advocacy association for the global redistribution of resources and power: Global Redistribution Advocates. We all want humanity to take charge of its own

destiny and organize itself to ensure that everyone has the basic necessities of life.

In a survey based on a representative sample of 499 French people that I conducted in the fall of 2016, only 1 percent answered *No* to the question: *Do you agree or disagree with the following statement? "I want humans to secure to all the conditions necessary for well-being: access to drinking water, food, healthcare, a healthy environment, safety, housing, education, information."* Building on this common goal, we must work together and overcome our disagreements.

To the best of my knowledge, no political party, association, or think tank has ever proposed a specific plan for global redistribution measures, consisting of a sizeable transfer of wealth from the rich to the poor at the international level, but, when I met with various political leaders and associations, I realized that they received these proposals with interest and even enthusiasm. It seems to me that if current political proposals lack a global vision, it is because existing structures lead to a blinkered approach. Indeed, the institutions that structure political debate, such as voting and the media, operate mainly on a national scale. This national structure clearly only favors national measures and organizations. Another obstacle to global action is the tendency of individuals to cherish their families and friends, but to feel powerless and lacking the necessary skills to take action on a wider scale. Unfortunately, problems such as climate change and malnutrition can only be solved by taking action on a planetary scale.

Our motivation to act for humanity is often bogged down by routine, personal worries or the inertia of social structures. But the most unbearable scourge holding us back is defeatism. It is an argument repeated endlessly: "I agree with you, but people are too indifferent/selfish/skeptical/indoctrinated; leaders will never lead such a reform." This reaction is the only one truly opposed to humanist proposals such as those presented in this book. In reality, people are willing to embrace ecological

change and solidarity, provided that the effort is international and equitably shared, and that it falls first and foremost on the wealthiest.

The problem with defeatism is not that it is *false*: we can have real reasons to believe that humanity is incapable of undertaking certain changes indispensable to its well-being; the problem with defeatism is that it is *bad* for us. If we discover an obstacle on our road to happiness, we must not give up, but find mechanisms to remove it. Human beings accomplish wonders through perseverance, and self-censorship out of pessimism is the first step on the road to decline.

Defeatism is a pessimistic prediction of the future with all its self-fulfilling aspects. It is an intellectual attitude which consists in thinking that, since the origin of life, predation has been the rule, and that it will continue to be so. It is thinking that society will reproduce its injustices and disasters. It is predicting the future as the arrival of what seems most likely to happen: what already exists. This position is not only scientifically inaccurate: we are not in a position to predict the future of humanity; above all, it is morally very serious. We must not predict our future, we must choose it. We must therefore combine our efforts to maximize our chances of experiencing a future that satisfies our desires. In contrast to predicting the most likely future, we must choose the best possible future. But for this best future to take place, it is not enough to just lovingly dream about it; we need to understand the mechanisms by which we can bring it about and the actions by which we can achieve it. In short, we need to build a project so that our future is in line with our hopes. Within this project, I believe it is necessary to include a Global Plan to end climate change and extreme poverty.

Rosalie, 2025

Rosalie was already awake when the roosters started crowing at 4am, threatening to rouse many of the dreaming inhabitants of the town of Houndé, in Burkina Faso. With a busy day ahead of her, she wrapped her shawl around herself and her newborn baby to protect both of them from the dewy mist of the early morning. It was a long walk up to the millet fields, but she didn't mind it. The few kilometres passed quickly and it did not seem long before she could spot the leafy majesty of the Senegal mahogany tree right next to her plot. There she could carefully place her tiny child and cover him with her shawl, in case it started to rain later. It was sowing season, and there was a busy day of hard work ahead of her. She bent low to alternate between digging the soil deep enough and carefully planting the tiny pearl-like seeds about a foot down. With time, each one would sprout into tall columns of the nutritious cereal. This year, she wasn't able to plant at the far end of the field, as the soil had become hard as rock; much too dry to cultivate.

At noon, when the sun was at its zenith, she sat with her baby boy, in the shade and ate the millet cake topped with shea butter, that her daughter had prepared the evening before. As she rested a little from the morning's toil, she took the time to breastfeed her son, who had also got an appetite for lunch it seemed. Nursing his precious head against her, the memories flooded back of the two young children she so tragically lost so early in their lives: her eldest, Saidou, who died of lung disease

after just six months of work in the Kiéné manganese mine, and her third child, Minata, who died of malaria at just five years old. She brushed away her tears hastily and went back to her sowing.

Shortly after sunset, around 8pm, Rosalie packed up her things into her tool bag, lifted the baby into her arms and set out for home. On the way, she met her neighbor, who greeted her with bad news. Two of his seven goats had died last month after drinking downstream from the gold-mining area. She made a mental note that she should avoid this cyanide-and mercury-contaminated water when she next goes gold panning during the dry season. Having reached home, Rosalie sat around the table with her husband and six children: for the evening meal, the family shared two eggs in addition to the daily millet paste tô with okra sauce. After dinner, Rosalie used the water can to wash the dishes and do her ablutions, while her eldest daughter studied by the light of their oil lamp. Rosalie was still feeling hungry but proud that the family had been able to buy her daughter a textbook by selling some of their millet reserves. As she finally lay down on her straw mat on the dirt floor, her back felt like it had been sawn in two; she shared the same pain that all the field workers suffered. She was looking forward though to next week, when she would be harvesting shea kernels with the women of her neighborhood. Not only was it much more fun than sowing, but with the sale of the almonds, she would be able to buy a third hen and some dried fish. While her youngest son danced in the street with his friends to the sound of the radio, Rosalie drifted off gently to sleep, not even hearing the song touting the virtues of water purification in a joyful mix of Mossi, Jula and French.

Chapter 1

An unbearable status quo

Major scourges afflict humanity. In this book, I focus on two of them: climate change and extreme poverty. The slow pace of progress in these areas is a disgrace for our society, which seems to care little for the vulnerable or for future generations. The situation is unbearable.

Climate change

The climate is a complex system, but the work of the IPCC has shown that we can approximate its evolution with a simple rule: the rise in global temperature is proportional to the cumulative CO₂ emissions since the industrial revolution.¹ To put an end to global warming due to the accumulation of CO₂ in the atmosphere, we need to achieve carbon neutrality. In other words, we need to bring CO₂ emissions down to zero. More precisely, emissions must reach *net* zero, insofar as residual emissions can be offset by equivalent capture through reforestation or artificial carbon sequestration. The temperature at which humanity

¹Figure SPM.10 in [IPCC \(2021\)](#) shows that one degree more corresponds to 2,000 GtCO₂.

chooses to stabilize the climate determines the carbon budget, i.e. the emissions we have left to emit. For example, from 2024 onwards, humanity only has a budget of 1,000 billion tonnes (Gt) of CO₂ to have two chances in three² to contain warming to +2 °C. To meet this carbon budget, CO₂ emissions (currently 38 Gt per year) could be reduced by the same number of tonnes each year, until they reach zero in 2077.

If, on the contrary, emissions continue to rise, warming could reach +4 °C by 2100, and up to +7-8 °C between the years 2300 and 5000.³ Antarctic melting could raise sea levels by 15 meters by 2500 and submerge coastal areas where 340 million people currently live by 2100.⁴ Vast areas of China, South Asia and the Middle East would be rendered uninhabitable in the 22nd century due to a lethal combination of heat and humidity.⁵ Even in a less extreme emissions scenario, with a temperature of +2 °C in 2100, sea levels would (in the absence of dykes) submerge areas where 190 million people currently live. If nothing is done to combat climate change, the population exposed to drought would triple by mid-century; this exposure being particularly pronounced around the Mediterranean and in the Middle East.⁶ In addition, 4 billion more people would be exposed to malaria and dengue fever, as the mosquitoes that carry these diseases move up into previously temperate zones, including Europe and Northeastern U.S.⁷ It is estimated that the increase in warming due to the emissions of a dozen Europeans or three Americans during their lifetime causes one additional death.⁸ Limiting

²Cf. Table SPM.2 in *IPCC (2021)*. The use of a probability (“two chances in three”) comes from the fact that climate models include a margin of error on the temperature reached by a given carbon budget.

³Montenegro et al. (2007).

⁴DeConto & Pollard (2016); Kopp et al. (2017); Kulp & Strauss (2019). In addition, areas home to nearly a billion people would be submerged by 2300.

⁵Im et al. (2017); Kang & Eltahir (2018); Pal & Eltahir (2016).

⁶Elliott et al. (2014); Marzi et al. (2021).

⁷Colón-González et al. (2021).

⁸Bressler (2021).

global warming to 2 °C would prevent 6 million annual deaths due to climate change by 2100 (as many as cancer deaths today).⁹ In the absence of adaptation measures, climate change will lead to a drop in agricultural yields of around 20 percent by mid-century for the main crops in sub-Saharan Africa. A warming of 2 or 3 °C, on the other hand, would lead to a drop in agricultural yields against which even adaptation measures would be ineffective.¹⁰ Generally speaking, our settlements, land uses and infrastructures are adapted to the current climate. Climate change will render many of them obsolete, if not simply destroyed.

To sum up, continued greenhouse gas emissions would jeopardize many parts of society, multiplying droughts and floods, reducing agricultural yields, increasing the likelihood of violent conflict, and leading to major population displacements.¹¹

Extreme poverty

The World Bank defines extreme poverty as consumption of less than €2 per day (adjusted for the cost of living). The €2 threshold (\$2.15 in constant 2017 dollars, to be exact) is expressed in purchasing power parity: it corresponds to what \$2.15 can buy in the United States. In a country like India, it takes less than \$1 to buy products that cost \$2.15 in the United States. Throughout the book, I use \$ (in italics) to denote the dollar in purchasing power parity (PPP) terms, and \$ to denote the nominal dollar. Similarly, I will denote the euro in PPP terms by € and the nominal euro by €.

This threshold is just sufficient to meet minimum nutritional needs. It's calculated that, in low-income countries, the extreme

⁹Carleton et al. (2022).

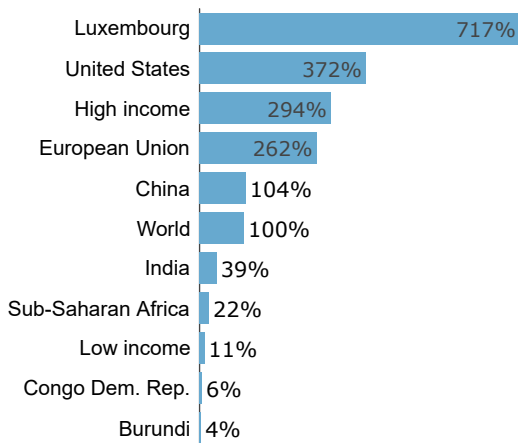
¹⁰Moore et al. (2017); Schlenker & Lobell (2010).

¹¹This paragraph repeats elements of the preamble to my Ph.D. thesis (Fabre 2020), and is based on numerous works (Carleton & Hsiang 2016; Cattaneo et al. 2019; Dell et al. 2012).

poverty threshold allows us to afford 3 m² in a home heated to 15 °C and a diet consisting solely of oil and a cereal (sometimes supplemented by lentils), which provides a daily intake of 2100 kcalories, 50 g of protein and 34 g of fat.¹² The number of people living in extreme poverty overlaps with the 700 million undernourished people in the world.¹³

Although the proportion of people living on less than €2 a day has been divided by four in the last thirty years, extreme poverty still affects two-thirds of the population in a country like Malawi. In fact, with population growth, there are more Africans living in extreme poverty today than there were thirty years ago. If extreme poverty has been reduced over the period, it is only thanks to the development of Asia, and China in particular.

FIGURE 1.1: GDP per capita relative to the world average, adjusted for cost of living (2022, World Bank).



China's GDP per capita is now around the world average,

¹² Allen (2017).

¹³ FAO (2023), World Bank.

at €1,000 per month.¹⁴ In comparison, GDP per capita is three times higher in high-income countries, like the United States or the UK and ten times lower in low-income countries such as Ethiopia or D.R. Congo (Figure 1.1). It is hard to exaggerate the gap in living standards between countries. In fact, a transfer of just 1 percent of GDP from high-income countries would double average income in low-income countries, home to 700 million people.¹⁵

The link between climate and poverty

Anyone who cares about the well-being of human beings wants to end worldwide poverty. Similarly, it is not necessary to attach any intrinsic value to nature or biodiversity in order to fight climate change; it is sufficient to be concerned about human well-being. Right now, climate change is already jeopardizing large sections of the population, and threatens all future generations, particularly in tropical countries. Indeed, global warming is all the more problematic in areas that are already hot, and home to most of the world's poor, as these areas are more exposed to drought, lower agricultural yields, and the difficulty of working outdoors (or without air conditioning). While the poorest are bearing the brunt of climate change, they also lack the means to cope with it: they don't have the resources to buy an air conditioner, build a dyke or migrate to an unscathed area. Not only does poverty increase vulnerability to climate change, climate change also increases the risk of poverty. By as early as 2030 it is estimated that between 32 and 132 million people will fall into extreme poverty as a result of climate change; particularly through its effects on health and agricultural commodity

¹⁴Nominal GDP per capita according to [World Bank \(2022\)](#).

¹⁵Of the 26 low-income countries, defined by GDP per capita below \$1,135/year, 22 are in sub-Saharan Africa and 4 outside (Afghanistan, North Korea, Syria and Yemen). 1.2 billion people live in a high-income country (with a GDP/capita of over \$13,845/year).

prices.¹⁶ By reducing growth in the poorest countries, climate change increases inequality: it is estimated that the income gap between the richest and poorest countries has already increased by 25 percent as a result of climate change.¹⁷

Climate change raises the question of the international and generational distribution of power and wealth. Indeed, the distribution of greenhouse gas emissions is extremely uneven: while the richest 1 percent of U.S. Americans emit an average of 318 t CO₂e per year, the average Indian emits 2 t and the poorest 10 percent of Rwandans emit just 0.1 t. This will be familiar to most as the carbon footprint, i.e. the emissions attributable to individual consumption, both direct, such as gasoline consumption and indirect in the emissions involved in producing the products consumed. Globally, the top 1 percent are responsible for 50 percent more emissions than the bottom half of humanity.¹⁸ And unlike many poor Africans or South Asians who have yet to be born; wealthy, elderly Westerners with large carbon footprints are unlikely to suffer much from climate change, and therefore have little interest in changing their opulent lifestyles, unless they care about the well-being of their descendants or humans in general.¹⁹ So, to prevent the dramatic impacts of climate change, it is misleading to frame the question simply in terms of temperature. The heart of the problem lies in the inequalities between human beings who differ in terms of wealth, location or generation. As such, a solution to climate change and its impacts can only be coherent if it is equitable, and therefore involves a substantial transfer of resources from today's rich to tomorrow's poor.

¹⁶Jafino et al. (2020).

¹⁷Diffenbaugh & Burke (2019); Khalfan et al. (2023).

¹⁸Bruckner et al. (2022); Chancel & Piketty (2015).

¹⁹As shown by vulnerability indices (Chen et al. 2015) or estimates of climate change damage as a function of country GDP (Burke et al. 2015).

Rajesh, 2024

Rajesh lay quietly on the floor in the corner of his sombre room. As the evening brought encroaching darkness and a very slight warm breeze through the open window, tears began to well up in his jet black eyes. His hard makeshift bed; a crude plastic mat, covered with a bedsheet given to him by his mother, provided little comfort tonight. He listened to the bustling outside from the closing flower market in the Mullick Ghat area; jutting up against the majestic towering Howrah bridge in Kolkata, West Bengal. He was quiet, but the city never was.

Flowing back to him, as if in a dream, he thought about his other life, his happy life. A carefree childhood wonder on his family's rice farm. Sunshine that lasted forever, coupled with uninterrupted joy and laughter. His sister Rani, two years younger than him, would beat him to the first upper branches of one of their three most treasured trees, eager for the sweet jewels of the prize. Large juicy ripe mangoes awaited them; red, gold and green, swinging heavily, from the efforts of their climb. Rajesh, bit away a piece of the skin and peeled carefully, then sunk his grinning face into the light orange pulp. Soon he was joining Rani in her giggles, as they sat there, faces and arms dripping with sticky sweet mango juice. He recalled his mother telling him the name: Himsagar. It sounded like the name of a prince, but it meant 'frosted sea'.

How he longed for the cooling respite of an icy glacier now, in the oppressive, heavily polluted heat of the Indian night. His

hot tears flowed easily.

His father and mother both worked long days on the small farm they owned in Burdwan. They kept a few chickens, and the rice they sold was always enough to get by with more than a little extra. But as Rajesh reached his teenage years, he learned to recognize a growing sadness in his father's eyes. The much-needed rains for the rice plants had become less reliable, and the extra pumping and irrigation required was becoming an expensive burden on the family's budget. At fifteen, Rajesh had no choice but to leave everything he loved behind, to find work in the city.

So now, he was here, in the employment of his uncle, and living in his home. But this was not his own loving family. The brother of Rajesh's father was a cruel, selfish man; indifferent to his younger brother, the struggling farmer. Business was good in Kolkata, selling wood to his many customers, delivered by his hard-working nephew Rajesh and a few other men. It was often back-breaking work. The larger timber consignments were always delivered by truck, but where the streets narrowed, or were simply mud pathways, Rajesh had to rely upon a rickety old trolley, or even haul the wood by hand. Being new to the city, Rajesh used to get lost, and getting lost meant losing time and making late deliveries. The day started at 5am and he would work into the early evening, until he felt he was ready to drop from exhaustion. He was always frightened that one day his uncle would dismiss him from the job, but at least, for now, he was able to send half his meager earnings back to his family. His family; he missed Rani so, so much. His tears tonight were not only for himself, but for her too. That night, he dreamed of a better life for all of them.

Chapter 2

The need for global redistribution

A moral imperative

Whether religious, philosophical or intuitive, morality prescribes transfers from high-income earners to low-income earners; any compassionate human being considers helping the poor of this world to be a necessary action and by extension the transfer of wealth from high-income countries to low-income countries. This is the case with utilitarianism, the reference ethical theory used in economics. Utilitarianism attributes the same weight to each person, and thus justifies the transfer of one euro from a rich person to a poor one, since one euro will bring more satisfaction to the latter. According to the theory of optimal taxation, this reasoning is valid as long as an increase in taxes does not encourage the richest to expatriate, conceal or reduce their activity to the point of diminishing the revenues obtained. Economists have calculated the optimal tax system taking these effects into account.¹ The theory of optimal taxation

¹In these calculations, [Kopczuk et al. \(2005\)](#) assume a single tax rate (a *flat tax*) and do not allow for a progressive tax schedule. Without this restriction,

can only rationalize the current situation by turning morality on its head. Indeed, researchers have shown that the currently very low level of international transfers is only optimal if we assign a value 2,000 times higher to a U.S. American than to a Congolese (or, alternatively, if we assign a value 100 times higher to the U.S. American and consider that only one-twentieth of the money transferred will reach its recipient, the rest being diverted by corruption).

A legal-diplomatic commitment

Beyond ethical considerations, the imperative of global redistribution has legal foundations. In 2015, all countries adopted the Sustainable Development Goals (SDGs), the first of which is the elimination of extreme poverty by 2030. Yet low-income countries do not have sufficient domestic resources to eliminate extreme poverty. Indeed, in the poorest countries, even taxing all income above 7€ a day at a 100% rate would not be enough to finance sufficient transfers to lift their entire population above 2€ a day by 2030.² In other words, it is impossible to achieve the first SDG without international transfers. And this, while the first SDG is limited to ensuring an income barely sufficient to avoid hunger. The transfer needed for this first SDG corresponds to 0.1% of global GDP, or as much as is spent on pet food.³

To ensure a decent life, which guarantees access to water, sanitation, education, a health system and a minimum ability to move about and socialize, it is estimated that an income of at least €7 per day is required.⁴ 620 million people live in a country where GDP per capita is below this threshold, and where it is therefore strictly impossible to ensure a decent life

the true optimum would be even more redistributive.

²Cf. Fabre (2024).

³Cf. [grandviewresearch.com/industry-analysis/pet-food-industry](https://www.grandviewresearch.com/industry-analysis/pet-food-industry).

⁴Cf. Chapter 7.

for everyone by mobilizing domestic resources alone. In total, half of the world's population lives below that poverty line.⁵ Closing the gap between their income and this threshold would cost 2 percent of global GDP in 2030.⁶

In 1970, industrialized countries pledged to allocate 0.7% of their GDP to official development assistance, including 0.2% of GDP for the least developed countries. This commitment, renewed in 2005 and 2015, has never been fulfilled. More precisely, actual foreign aid amounts to only half of the aid promised, including only 0.06% of GDP for the least developed countries, and only a handful of countries are meeting their commitments: Luxembourg, Sweden, Norway, Germany and Denmark.⁷ It is estimated that the bulk of the Sustainable Development Goals could be achieved if industrialized countries finally honored this commitment.⁸ In order to achieve a maximalist version of the SDGs (including ensuring access to clean energy) or another ambitious target in relation to the status quo (such as ensuring €7 per day for everyone), high-income countries would need to transfer more resources, probably between 2 and 5 percent of their GDP.

An imperative for decarbonization in the South

Lastly, even someone who didn't necessarily even think about the climate, and was not bound by ethics or international commitments, and only cared about their own well-being and their descendants, should support a climate justice solution. To finally put an end to climate change, the whole world needs to

⁵Cf. ourworldindata.org.

⁶In purchasing power parity, this gap (the *poverty gap*) is \$4,800 billion, or 3.4% of global GDP of \$140,000 billion. Assuming annual world growth of 3.5% (the rate observed over the last 20 years), we find a poverty gap of 2.1% of world GDP in 2030.

⁷(OECD 2023).

⁸SDSN (2019).

decarbonize. However, international transfers are a prerequisite for low-income countries to decarbonize. On the one hand, these countries have other priorities than decarbonization, and therefore deploy the most affordable energy system, often based on coal. On the other hand, these countries argue, and rightly so, that they are the most vulnerable to climate change and have contributed only marginally to it.⁹ In international negotiations, these countries generally announce two emission reduction targets: an unconditional unambitious target and an ambitious target conditional on external financing. For example, Ethiopia has pledged to unconditionally reduce its emissions by 14 percent by 2030 compared with a no-climate-action scenario, and is making a 69 percent reduction conditional on \$250 billion in financing. At the same time, developing countries, supported by the UN General Secretariat, are calling for annual transfers of \$100 billion to compensate for the loss and damage caused by emissions from developed countries.¹⁰ While experts estimate that the transfers required would be two to four times this amount, only \$700 million were *promised* in 2023 at COP 28.¹¹

More generally, the international arena is tending to polarize between the Global South, which is demanding a fairer world order, and the Global North, which is reluctant to give up its dominant position. For example, the African Union recently called for a global carbon pricing regime, a tax on financial transactions and a reform of the financial system, in order to benefit from dedicated, affordable and sustainable climate financing, dissociated from national and geopolitical interests.¹² Global South countries are also critical of international negotiations on corporate taxation, which are taking place under the aegis of the

⁹Africa and South Asia combined are responsible for 6 percent of cumulative CO₂ emissions.

¹⁰Cf. SGNU (2023); TC (2023).

¹¹Cf. Markandya & González-Eguino (2019); Robinson et al. (2021); Songwe et al. (2023) and www.theguardian.com/environment/2023/dec/06/700m-pledged-to-loss-and-damage-fund-cop28-covers-less-than-02-percent-needed.

¹²African Union (2023).

OECD and exclude many low-income countries. In November 2023, although the OECD countries voted against it, the Global South countries pushed through a resolution at the UN establishing a Convention on International Tax Cooperation (akin to the climate COPs). Global South countries hope that this negotiating framework will be more favorable to their interests than the OECD.

Despite the tendency of countries in the Global North to defend their short-term financial interests, there is hope that they will acquiesce to certain demands from the Global South. Whether in defense of humanist values or to restore its often tarnished image, the West seeks to present itself as a model based on human rights, democracy and sustainable development. For example, Ursula von der Leyen, President of the European Commission, recently came out in favor of global carbon pricing,¹³ and a working group on international taxation (including carbon) was launched in December 2023 jointly by France, Kenya, and Barbados.¹⁴

Humanist values would at last be properly defended in the event of an international climate and fair taxation agreement that would make the Global North pay. Such an agreement would have the potential to rebuild geopolitics on solid foundations and pacify international relations. In contrast, climate change has been shown to increase the risk of armed conflict, particularly in Africa.¹⁵

In subsequent chapters, I will propose a Global Plan to end climate change and extreme poverty, involving major North-to-South transfers, accepted by a majority of the populations of Global North countries.

¹³Cf. her [Tweet](#) of September 9, 2023.

¹⁴Cf. https://bit.ly/taskforce_tax.

¹⁵Burke et al. (2009); Eberle et al. (2020).

Catherine, 2025

Catherine had just finished her long ten hour overnight shift at the hospital in Poitiers. Her work pattern had radically changed since the pandemic. With so many of her colleagues wanting to leave, or having already left, staffing levels were putting a strain on all who remained and in spite of a moderate recent pay rise, morale among the nurses remained low. As she walked out to her faithful old, slightly beaten-up little Renault Twingo, it was already starting to rain. She hastened her pace, reminding herself, with a small sigh, that she wasn't the quitting type, and that she'd already been through far worse and survived that. After some sleep today, she would be able to pick up her young daughter Amélie from school in person, rather than relying on mamie to do it. Amélie was already proving to be her mother's daughter; she'd been struggling in mathematics, but just like her Mom, she wasn't a quitter. Catherine was looking forward to having some precious time with her later, and that thought alone was enough to raise her spirits during the easy twenty minute drive back home to Charassé.

When the divorce had seemed inevitable, Catherine had been forced to give up their modest two bedroom apartment near to the hospital, and move back home. As if that wasn't enough of a shock, less than a year later, she lost her father to a stroke. It was so sudden and unexpected and devastating. A friend had told her "Cathy, always remember, when things look bad, your life can change in thirty minutes; you never know

what is round the next corner!" It was supposed to be a pep talk, but now she felt that life had been particularly cruel to her, and she was yearning for the next 'corner' to bring something happier.

As she pulled up outside her family home, she knew her mother would be there with that big bowl of hot chocolate she'd always had as a kid. It was reassuring to return to the old familiar place where she could forget about the demands of work, for a few hours at least. Sure enough, Simonne was there in the kitchen, already reaching into the cupboard for the breakfast things. She greeted her mother with a hug from behind, a long, slow grateful embrace and placed a tender kiss on the back of her neck. Simonne, busy with dishes and cups, hunched up slightly and briefly, to acknowledge her daughter's affection. After breakfast, and a catch up with Mom, a shower and bed awaited her. As she blew gently, then cautiously sipped directly from the steaming bowl of delicious comforting sweetness, she really did believe her friend's advice. Things really could change in just thirty minutes, as long as you just kept going, and did not give up, when life dealt you an unexpected blow.

Later that day, Catherine stood, leaning slightly on her car, waiting patiently for a glimpse of her daughter's distinctive bright green coat, as she emerged from her school's main entrance door. The rain had finally stopped, and sunshine was randomly glinting on the puddles in the school yard. She watched her daughter skipping with joy towards her, jumping the puddles with careless abandon. She appeared to be tightly gripping a paper, waving in her hand and she was positively beaming. Catherine threw open her arms wide in anticipation of the big hug that was coming her way. Her daughter was so excited, that she didn't even wait for her Mom to question her about the school day. "Mommy!" she exclaimed, "Madame Lambert gave me an 'A' in Mathematics! Plus, plus!" she was practically bursting; "We learned about the Earth today!" They both got into the car, as Amélie continued; "We have to fix it Mommy!"

We have to fix it! That's what I'm going to do. I'm gonna help you Mommy, and then I'm gonna help the Earth!"

Chapter 3

The heart of the Global Climate Plan

We saw in Chapter 1 that there is a carbon budget that must not be exceeded to keep global warming below a given target. The Paris Agreement sets this target. Signed by all countries in 2015, it aims to contain warming “well below 2 °C (...) and to pursue efforts to limit the temperature increase to 1.5 °C”.

How can we guarantee an emissions trajectory in line with this carbon budget?

The safest approach would be to cap global emissions, with an annual **cap** that decreases in line with the target.

How can we make sure that the sum of all emissions do not exceed the cap?

Emissions permits corresponding to the annual quota would be auctioned to polluting companies at the source of carbon emissions, such as fossil fuel producers or importers.

How then to allocate the revenues from this carbon price?

The simplest is to **share** them equally among every human.

The answers to these three questions form the heart of the Global Climate Plan.

Preventing misunderstandings It is important to understand that by introducing a cap on emissions, global emissions are by construction equal to this cap, thanks to mechanisms (detailed in Chapter 5, pp. 54 and 62) that prevent it from being exceeded. This is the main advantage of this Plan.

This cap settles the temperature issue, but not the social question: It is the distribution of the carbon price revenues that determines how efforts are shared. The egalitarian distribution I propose will bring about a major North-to-South redistribution and end extreme poverty.

To avoid common confusion, it should be noted from the outset that this system would not allow carbon offset credits to be purchased through reforestation projects. The proposed system resembles the European carbon market in place since 2005, but is radically different from the dysfunctional and greenwashing carbon offset market. The European market caps emissions from the industry and electricity sectors, on a trajectory that decreases to zero by 2040. It will be supplemented in 2027 by a second carbon market covering emissions from transport and buildings. These carbon markets should not be confused with the carbon offset market, which allows companies or individuals to buy carbon credits to offset their emissions through projects (such as reforestation) in developing countries. Carbon offsetting is problematic because it does not guarantee a reduction in global emissions. Indeed, it is not clear whether the projects financed result in a reduction in emissions that is *permanent* (for example, an area reforested during the project may be deforested later) and *additional* (for example, a forest could have grown back even in the absence of financing, or reforestation in one place may generate *carbon leakage* by inducing deforestation in another). For these reasons, carbon credits are not eligible on Eu-

ropean carbon markets. Carbon markets truly reduce European emissions.

Functioning The system outlined above can be implemented as follows. Each year, a limited number of emissions permits are created, in line with the emissions trajectory set. These emission permits are auctioned to the companies at the source of the CO₂ emissions, and in particular to those who market coal, oil or gas. These companies must purchase permits corresponding to their emissions. Finally, the revenue generated by the sale of permits is redistributed as an equal basic income for all human beings.

Who pays? Who receives? The basic income is equal to the revenue generated divided by the population. The revenue generated is equal to the carbon price multiplied by global carbon emissions. Thus, the basic income is equal to the carbon price multiplied by the average human carbon footprint.

Although, for practical reasons, we would distribute money rather than emission permits to individuals, these two options are equivalent. Let's imagine that every human being received the same emission permit and could sell it to polluting companies on the carbon market. The amount he or she would receive would be equal to the carbon price multiplied by the individual quota, i.e. exactly the basic income. In this way, there is a perfect match between egalitarian allocation of emissions permits and egalitarian allocation of revenues.

Even if some would like to believe that it is the polluting companies paying the price for their pollution, in reality, it is the consumer who bears the cost. This is because polluting companies pass on the cost of emission permits in the form of higher prices, so that each individual faces a higher expense, equal to the carbon price multiplied by their carbon footprint. As a result, the basic income exactly covers the carbon price paid for an individual whose carbon footprint is equal to the world average. Individuals with a carbon footprint higher than

the world average lose purchasing power¹. Conversely, people with a low carbon footprint gain. In other words, the proposed system operates a global redistribution from the polluters to the frugal, so, to a first approximation, from the rich to the poor. A recurring argument against carbon pricing is that it would give the richest a free pass to pollute, and shift the cost of decarbonization onto the middle classes, who can't afford the price rises.

The short answer to this objection is that we need to be clear about what we are comparing ourselves to. If we compare the Global Climate Plan with a far more radical proposal, such as a way out of capitalism where incomes would be capped at €3,000 per month, then the richest people are indeed doing well. On the other hand, compared to the status quo, the richest lose out, and all the more so when we implement the complementary measures described in Chapter 7.

For the long answer, it is best to start by recalling the Plan's distributional effects: individuals with a high carbon footprint would lose out financially, while individuals with a carbon footprint around the world average could make the ecological transition without losing purchasing power. These "average" individuals would be incentivized to change their equipment and habits by the carbon price, and those who switched to decarbonized options faster than others would benefit financially, as their carbon footprint would fall below the global average. On the other hand, the middle classes in high-income countries have a higher carbon footprint than the world average. They would therefore lose out financially if they did not adapt and, in any case, adaptation itself presents a cost in terms of money or comfort levels. However, the better-off would lose even more, since they have an even higher carbon footprint.

Of course, it is fair to say that the better off have more leeway

¹This result stems from assumptions justified in Appendix B. The carbon footprint and its global average mentioned are those *after* the Plan comes into force.

to adapt, and should be asked to contribute even more than the Plan implies. For this reason that I advocate additional measures of national redistribution (cf. Chapter 7) to ensure that the full cost of decarbonization is borne by the most affluent, and to preserve the standard of living of the middle classes in high-income countries. However, if the majority of the population supports national redistribution measures, it is because they consider the distribution of wealth too unequal, whether there is a climate policy or not. In other words, we can separate the two proposals: on the one hand, a climate policy that does not aggravate inequalities (in fact, the Global Plan reduces them); on the other, a national redistribution policy that puts an end to an indecent level of inequality. If we were looking for a complete political program, we would have to include these two proposals, as well as many others. But that is not the purpose of this book. Here we focus on one proposal, and design it in such a way that it can be supported by individuals and governments of all stripes.

To sum up, I believe that the Global Climate Plan is preferable to the status quo, and should be complemented by additional redistributive measures.

Carbon pricing or rationing? Setting up a carbon price is preferable to a system of non-tradable individual carbon quotas for several reasons, not least of which is that the market is more favorable to the poorest, as it provides financial compensation to those who do not use up all their quota. For a detailed argumentation, see p. 113.

Tax or quota? A carbon market sets a quantity: the regulator sets an emissions quota and lets the market determine the carbon price. Conversely, we could imagine a carbon tax: the regulator sets the price and lets the market determine emissions. Provided that the carbon price is the same in the system that sets the quantity and the one that sets the price, the two systems

are strictly equivalent (they result in the same emissions priced at the same level). The proposal is based on a system that fixes the quantity, since the primary objective is to respect the carbon budget (and fixing the price does not make it possible to forecast emissions accurately). The advantage of a carbon market is that it places a cap on emissions, ensuring that the desired emissions trajectory is followed. If the trajectory of the tax is set in advance, there is a good chance that the level intended to achieve the climate target will be wrong, and that the tax will be set either too low or too high. If the level of the tax is automatically re-evaluated each year so as to keep emissions on the desired trajectory, we end up with a system rather like the carbon market. The difference between the two is one of detail: the market adjusts immediately to new situations, such as war or the entry of a new country into the system, at the cost of an ecosystem of financial players who analyze, operate and earn a return on this market. As the two systems are very similar, an automatically reassessed tax would be just as appropriate. However, I prefer to present the Plan in the form of a market, so as not to be confused with the usual carbon taxes, whose trajectory is fixed in advance.

Finally, if the level of the tax were re-evaluated each year by a political decision, rather than automatically, this would offer interest groups endless opportunities to question the level of climate ambition and deviate from the climate target. To better resist such pressures, it seems to me preferable to set the emissions trajectory in advance. This preference means prioritizing the climate objective, even if it means accepting major reductions in living standards if decarbonization proves more costly than expected. In contrast, some would probably prefer to limit efforts in the short term, even if it means inflicting more damage on future generations if efforts prove less effective than expected. If this latter inclination were in the majority, the climate objective could no longer be guaranteed. We would then have to resolve to have a price ceiling on the carbon market, or

a carbon tax whose level would be chosen politically.²

Conclusion To sum up, we can put an end to global warming by capping emissions, and eliminate extreme poverty with a basic income. A simple and effective system for dealing with both problems is to combine these two solutions. This is the heart of the Global Climate Plan, which constitute the first two principles detailed in Chapter 5. I will also explain how systems of justice and geopolitics will come into play to complete the plan in Chapter 5. Finally, this Global Climate Plan must be complemented by a series of other measures: I outline them in Chapter 7.

²We could then set the tax level at the median of the preferred levels, as suggested by Weitzman (2017).